Land Cover Atlas of Afghanistan 2025Brief Technical Note

Methodological details | Comparison with FAO 2016 & NSIA updates | UN cartography | Land-area changes

Revised brief — aligned with FAO Geospatial Unit Q&A (WBK-EFSP AF2, 15 Dec 2025)Sources: FAO Land Cover Atlas of Afghanistan 2024 (published 2025); FAO 2016 Land Cover Atlas; NSIA updates

1. Methodological details of the 2025 FAO Land Cover Atlas

What is it?

Updated national land-cover atlas for Afghanistan (mapping year 2024; published by FAO in 2025). It provides a single-year, observation-based national baseline at national, provincial, district and river-basin levels.

Technology

- Cloud computing: Google Earth Engine (GEE).
- Classifier: Random Forest machine learning (GEE Python API).
- Hierarchical classification into 21 detailed classes (aggregated to 11 for atlas presentation).
- Automated irrigated vs rainfed separation using WaPOR evapotranspiration + CHIRPS precipitation (needs further validation).
- Multi-temporal Sentinel-2: 12 months (Oct 2023–Sep 2024), with four-month median composites to reduce cloud effects.

Software / tools

- Google Earth Engine (Python/JavaScript)
- Google Earth Pro and Collect Earth Online (CEO) for sample review
- UN Geospatial / NSIA boundary datasets
- FAO Agro-informatics Platform for digital outputs

Resolution / granularity

- 2024 land cover map: 10 m (Sentinel-2).
- Supporting layers: DEM 30 m; WaPOR ~300 m; CHIRPS ~5.5 km (smoothed) for irrigated/rainfed assistance.
- Reporting units: 34 provinces, 422 districts, 8 regions, 5 river basins / 36 watersheds.

Legend / standards

Based on FAO LCML/LCCS. Eleven aggregated classes: Built-up; Fruit trees; Vineyards; Irrigated herbaceous cropland; Rainfed agricultural land; Forest and shrubland; Rangeland; Bare areas; Sand cover; Waterbodies and marshland; Snow cover.

Total irrigated land = irrigated herbaceous cropland + fruit trees + vineyards (includes both seasonal and permanent irrigation).

Training data and accuracy

- ~105,000 training points collected — the largest national land-cover training set used for this product; covers irrigated and rainfed classes among 21 classes.
- Accuracy assessment uses an independent hold-out (reported as ~30% of collected points / 26,115 test points in atlas tables).
- Overall accuracy 0.872 (87.2%); Kappa 0.858.
- Lowest F1 ≈ 0.78 (non-urban built-up), still aligned with literature; most classes F1 > 0.85.
- Area confidence intervals at 95% reflect class performance (F1-informed) for min/max expected areas.

Key 2024 area figures (agreed FAO answers)

- Total land area to use: 64.18 million ha (UN Geodata / UN Geohub) — atlas to be reviewed accordingly.
- Total rainfed agricultural land: 3.07 million ha.
- Total irrigated agricultural land: 4.00 million ha (herbaceous + vineyards + fruit trees).

National expert involvement

National/FAO Afghanistan experts were involved in methodology definition, ground-data collection and result review/validation (acknowledged in the Atlas). Named contributors in the FAO Q&A include: Sanju Upadhyay, Sayed Sharif, Maziar Karimi, PuspaRaj Khanal, Muhammad Safi, Waheedullah Yousafi (FAO AFG). Further NSIA consultation is proposed without listing additional personal names.

2. Methodological comparison table (2016 FAO | NSIA updates | 2025 FAO)

NSIA updates are operational cultivated-land / crop statistics from satellite interpretation — not a full multi-class FAO-style land-cover atlas.

Methodological aspect	2016 FAO Land Cover Atlas	NSIA subsequent updates	2025 FAO Updated Atlas (2024 map)
Purpose	**National multi-class land-cover database/atlas (circa-2010; published 2016)**	**Seasonal cultivated area, irrigated/rainfed crop extents, yields**	**Updated national multi-class land-cover baseline for 2024**

Technology	FAO/GLCN object-based mapping + visual interpretation/enrichment	Remote sensing / satellite-image interpretation	GEE cloud ML (Random Forest); hierarchical classification; WaPOR+CHIRPS irrigated/rainfed tool
Software	FAO MADCAT / LCCS tools	National RS/statistical workflows (not published as full LC atlas)	Google Earth Engine; Google Earth Pro; Collect Earth Online
Main imagery	SPOT/Landsat era inputs; HR used mainly for points/validation	Satellite imagery for cultivation monitoring	Sentinel-2 time series (Oct 2023–Sep 2024) + DEM / climate auxiliaries
Spatial resolution	30 m land-cover product (HR only for sample validation)	Operational cultivated mapping (class set limited)	10 m land-cover product
Legend standard	FAO LCCS; ~25 classes → 11 aggregated	Focus on cultivated / irrigated / rainfed / crop-specific stats	FAO LCML/LCCS; 21–24 classes → 11 aggregated
Temporal basis	Multi-year / potential-oriented earlier assessment window	Annual / solar-year campaigns	Single-year observed snapshot (2023/24)
Training / validation	Visual interpretation + ancillary/VHR support	Operational image interpretation	~105,000 points; independent accuracy assessment
Accuracy reported	Expert QC / enrichment; no OA/Kappa package like 2025	Not published as full multi-class OA/Kappa atlas metrics	OA 87.2%; Kappa 0.858; UA/PA/F1 + confusion matrices
Total land area frame	National frame ~64–65 Mha depending on source	NSIA statistical area references	Agreed: 64.18 Mha (UN Geodata / UN Geohub)
Admin units	Provinces / districts / basins	Official NSIA admin reporting	34 provinces, 422 districts, 8 regions, 5 basins

Resolution note (FAO Q&A): 2016 LC is 30 m; 2024 LC is 10 m. This difference can affect comparative analysis and must be considered when interpreting change.

3. UN cartographic rules / procedures applied

(i) Demarcation of international boundary

- International boundaries / base layers prepared from UN Geodata (UN Geospatial / UN Geohub).
- Map legend distinguishes International boundary from provincial boundaries.
- Standard UN/FAO disclaimer applies to all map plates.

(ii) Contested / disputed international boundaries

- Legend includes Disputed boundary symbology (not treated as an ordinary undisputed international line).

- Narrative refers cautiously to the Jammu & Kashmir border context “as currently recognized in agreements between India and Pakistan,” consistent with UN status-neutral practice.

- Repeated note: “Refer to the disclaimer on page ii for the names and boundaries used in this map.”

(iii) Total land area of the country

- Agreed figure for atlas use: 64.18 million ha from UN Geodata (UN Geohub); atlas to be reviewed accordingly (FAO–WB Q&A, Dec 2025).

- Practice: internationally endorsed boundary geometry for cartography; agreed area total for statistics.

- Slight variations by source (e.g. NSIA / FAOSTAT) are acknowledged in the atlas text.

(iv) Sub-national / provincial and district boundaries

- Administrative boundaries from NSIA (provincial, district, regional shapefiles), used with UN geospatial references.

- Analysis/reporting for 34 provinces, 422 districts and 8 regions.

- Hydrological reporting uses 5 river basins / 36 watersheds (AIMS/HDX), separate from political boundaries.

Standard disclaimer text used: designations and presentation do not imply any opinion by the UN Secretariat on legal status or delimitation of frontiers/boundaries.

4. Land-area comparison table (major classes)

2016 figures from FAO 2016 atlas/comparative Table 6; 2024/2025 figures from FAO atlas + Dec 2025 agreed Q&A totals. NSIA column = closest public cultivated-land figures (not full LC classes).

Land class	2016 FAO Atlas	NSIA subsequent updates	2025 FAO Atlas (2024)	Change 2016→2024 (FAO)
Total land area frame	~64–65 Mha (source-dependent)	NSIA/FAOSTAT vary slightly by year	64.18 Mha (UN Geohub) — agreed	Frame harmonized to UN Geodata
Built-up	0.31 Mha (0.5%)	n/a as full LC class	0.7 Mha (1.0%)	Increase (≈ doubled)
Fruit trees	0.12 Mha (0.2%)	n/a	0.4 Mha (0.6%)	Strong increase
Vineyards	0.08 Mha (0.1%)	n/a	0.1 Mha (0.2%)	Slight increase
Irrigated herbaceous cropland	3.6 Mha (5.6%)	Irrigated cultivated land often cited ~2.1 Mha (seasonal cultivation definition)	3.5 Mha (5.4%)	Slight decrease (–0.12 Mha)
Total irrigated (herbaceous + fruit + vine)	5.9% of country	Irrigated farmland within cultivated stats (definition differs)	4.00 Mha (agreed Q&A); ~6.2%	Slight overall increase

Rainfed agricultural land	3.7 Mha (5.8%)	Rainfed cultivated often ~0.87–0.92 Mha in recent statements (in-season)	3.07 Mha (agreed Q&A); 4.8%	Decrease (–0.66 Mha)
Forest and shrubland	1.7 Mha (2.8%)	n/a	1.5 Mha (2.1–2.2%)	Slight decrease
Rangeland	30.2 Mha (47%)	n/a	28.6 Mha (44.4%)	Decrease (–1.7 Mha)
Bare areas	17.4 Mha (27%)	n/a	18.5 Mha (28.7%)	Increase
Sand cover	4.78 Mha (7.4%)	n/a	6.5 Mha (10.1%)	Increase
Waterbodies & marshland	1.85 Mha (2.9%)	n/a	1.2 Mha (1.8%)	Decrease
Snow cover	0.5 Mha (0.8%)	n/a	0.6 Mha (0.9%)	Slight increase

Why both rangeland and rainfed declined

Both depend on precipitation/climate; recurring drought over the last decade affected both (FAO Q&A). Not a simple conversion of rangeland into rainfed.

How to read NSIA vs FAO numbers

- NSIA figures are mainly in-season cultivated land; FAO classes are full land-cover masks — so NSIA cultivated totals are typically lower.
- For full landscape change (rangeland, bare, forest, built-up, water), use FAO 2016 vs 2024 comparison (atlas §5.2.3), with the 30 m vs 10 m caveat.

LC Atlas vs Agro-Ecological Zone (AEZ) Atlas

- LC Atlas = actual land cover picture for 2024.
- AEZ Atlas = land potential (potential yield/area/crop suitability).
- They are complementary, not substitutes.

5. Quick Q&A (aligned with FAO–WB briefing, Dec 2025)

Q: What total land area should be used?
A: 64.18 million ha (UN Geodata / UN Geohub). Atlas to be reviewed accordingly.

Q: Total rainfed / irrigated agricultural land (2024)?
A: Rainfed 3.07 Mha; Irrigated 4.00 Mha (herbaceous + vineyards + fruit trees).

Q: Is irrigation seasonal or permanent?
A: Both seasonal and permanent irrigation are included in irrigated area.

Q: Where is the land-cover change matrix?
A: Atlas §5.2.3 — class variation from 2016 to 2024.

Q: Was 2016 imagery higher resolution than 2024?

A: No. 2016 LC = 30 m; 2024 LC = 10 m. HR data in 2016 mainly for points/validation.

Q: How was accuracy assessed?

A: Independent hold-out from the ~105,000 collected points (not used in training); OA 87.2%, Kappa 0.858.

Q: Can more irrigated/rainfed training help?

A: Already very large sample; WaPOR+CHIRPS method exists but needs validation; dataset can later support crop-type upgrades.

Boundary representations follow UN/FAO disclaimers and do not imply determination of sovereignty or frontiers.

Primary references: FAO 2025 Land Cover Atlas of Afghanistan 2024 (DOI: 10.4060/cd3186en); FAO 2016 Afghanistan Land Cover Atlas; FAO Geospatial Unit Q&A for WBK-EFSP AF2 (15 Dec 2025).